

THE CRUMBLING MAINSTREAM RESISTANCE
TO THE PARANORMAL
or
NEW SCIENTIFIC CONFIRMATIONS REGARDING THE
EXISTENCE OF CERTAIN PSI FACULTIES
or
YOUR SEVENTEEN SENSES



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How many of you here today would like to have seventeen senses rather than just five of them? How many think that seventeen would be better than just five? How many of you here already know than you have more than five senses?

When Mohammed Ramadan and Clarence Robins asked me to come here and rap about something, we had a little difficulty deciding upon a topic worthy of your interest. Finally it dawned on me that there exists what we might call a Particular Situation regarding psychic perceptions. We might perhaps better grasp the nature of this Particular Situation if I ask you:

- (1) How many of you are vaguely or more deeply aware that parapsychology exists and what it deals with?
- (2) How many of you are vaguely or more deeply aware of the scientific non-acceptance of parapsychology?

Well, for those of you who know little of this, parapsychology deals with human psi faculties (such as ESP, telepathy, and clairvoyance), and does so more or less outside of mainstream scientific acceptability. I'll discuss the whole of this in what follows.

The Particular Situation I've already referred to is that on the leading edges of science during the last twenty years or so, a somewhat new interdisciplinary field has been coming into existence between parapsychology and the scientific rejection of it. This new field is not comprised of parapsychologists; but it is comprised of scientific disciplines completely accepted in science. Shortly, I'll risk boring you to distraction by identifying twenty-one of these cutting-edge fields. However, in this new interdisciplinary field the biological and organic basis for many forms of ESP have been conclusively discovered. All of these discoveries have been published in some form, scientific or popular, with commentary that the entire question of human psi performance should be completely and positively reevaluated by science, psychology, psychiatry, and society.

You would therefore think that the debate about whether ESP exists or not should be retired into history. Yet nothing of the kind has happened, and our society, its hard and soft sciences, its skeptics, its media, its academe, and its attitudes just go on as if nothing has taken place along these lines. Some of you here may already know which discoveries I'm referring to; but chances are that you may not be at all aware of them. I myself know many parapsychologists and scientists who are not even aware of them. In fact, about the only group that has taken adequate and accurate interest in these astonishing discoveries are the scriptwriters of the TV series Star Trek: The Next Generation.

So I decided to review this Particular Situation for you here today, and perhaps find out what you think about it, or how it can be explained more thoroughly.

But first, let me give you a little of my background. During the course of 1970, I began volunteering as a test subject in psi experiments in several New York parapsychology laboratories. Some of these experiments demonstrated unusual successes and subsequently, by 1972, a good deal of media attention. I was then invited to take part in experiments in a number of other laboratories, and in 1973 became part of a large effort at Stanford Research Institute International, the nation's second largest "think-tank" based in Menlo Park, California. I worked within these auspices, supported by on-going official funding, until I retired from active lab research in 1989.

The overall scope of the project at SRI presented unusual opportunities regarding what had been called psychic or parapsychology research, but which evolved there into novel concepts, such as psychoenergetics and bio-communications. These concepts were formulated on basic principles somewhat differing from those typical of parapsychology that had held sway from approximately the early 1930s. The on-going project at SRI ultimately came to involve over 500 participants, scientists, and serious over-sight officials, all collected into a broad multidisciplinary examining body which considered a large and diverse number of factors relating to human psi potentials.

During this 17-year period, though, and quite independently of the SRI and other parapsychology efforts, significant discoveries were being made in other scientific fields considerably outside of the scope of parapsychology. To the surprise of those non-parapsychological researchers, many of their discoveries were applicable to determining the real existence of certain psi or extrasensory faculties in the biological human. Our project began incorporating these new discoveries into our understanding. The meaningful implications of the sum of such discoveries is the topic of this talk.

Unfortunately, in order really to put all this into perspective, it seems necessary to first drag you through what might be called the anti-psi historical situation, since it is this situation that contains the reasons for earlier scientific and mainstream resistance to psi and against which the meanings of the new discoveries have to be contrasted. You can sleep through this part of this talk if you want; but in any event don't mistake it for an attack upon science or psychiatry or social values. I'm merely reviewing historical contexts that are pertinent to the new discoveries -- and to your seventeen senses that are ripe in potential.

So, to first give some historical perspective here, it is well-known that organized scientific resistance to all forms of psychical research has prevailed since at least 1882 when the first Psychical Research Society was founded in London both outside of and ultimately as an antagonistic and discredited counterpart to other modern-age scientific disciplines. The societal power and influence of modern science was absolutely enormous, and its precepts were concretely extended into academic and media collectives, and hence into what is often called the "mainstream."

As but one social-influencing example of this, when I first entered into psychical experiments in 1970, Time magazine, when it took note of parapsychology research and developments, did so by reviewing them in what was, in print, entitled "The Fraud Box." Such parapsychological luminaries as Dr. J. B. Rhine, and his wife and co-worker, Dr. Louisa Rhine, found themselves reviewed in the fraud box. It's worth noting that Time stopped featuring its fraud box only in about 1973 when it was learned that the Pentagon and the CIA had become interested in psychic matters, since it would have been slightly politically incorrect to feature either the Pentagon or the CIA in the Fraud Box.

The general rationale and specific reasons behind the rejection of psi during the modern scientific epoch have always been understood and openly stated. The reasons are both philosophical and scientific. Within the constructs of philosophical materialism that dominated modern-age scientific understanding and expectations, it was broadly held that psi faculties belong to superstition, and if not that, at least to the religious concepts of the soul, the religious concepts themselves being considered as unscientific in origin.

More basic to scientific concepts themselves, though, it was understood that the biological human possessed only five physical senses, and that none of these, as then known, could account for the existence of extraordinary perceptual faculties. The formal reason for scientific resistance to psi was that until a direct biological or organic basis for psi perceptions could be

discovered, hard scientists were obliged to ignore claims that they existed.

The absence of such bio-organic explanations for psi faculties was extended into the modern soft sciences of psychology and psychiatry as the basis for considering all claims of such faculties to be either of illusory, hallucinatory, or psychopathological etiology or cause. These attitudes against psi have been maintained for over a century, and are still maintained via many professional and mainstream concepts of them.

As but one example, I'm now going to refer to the third edition of the Diagnostic and Statistical Manual of Mental Disorders, published by the American Psychiatric Association in 1987. I'm referring to this Manual not as an antagonist to psychiatry, but because it serves as a general and authoritative professional reference book widely available in our cultural mainstream. The third edition is a revision of two earlier Manuals, themselves based in part on even earlier manuals generally grouped together as The International Classification of Diseases begun in France about 1900. The present edition of the International Classification is a function of the World Health Organization today, and thus a function of the United Nations complex of activities. The present 9th edition of the International Classification contains a Section V regarding the classification of mental disorders, and which was incorporated by the American Psychiatric Association into its 1987 Manual.

I'm obliged to point out here that the Manual is not only used by psychiatrists, but by psychologists, family doctors and counsellors, educators, sociologists, higher-education students, law enforcement officials, government officials, congressional persons, and, of course, various scientific disciplines. In my experience as an artist living in New York, I've even found it consulted by museum curators and gallery dealers.

Now, as I'll discuss ahead, because the American Psychiatric Association is about to make a startling and extraordinary change in its upcoming fourth revised edition of the Manual, its worth our understanding to take a very brief excursion into the pertinent part of the third edition.

If you consult that third edition of the Manual, you will find on pages 194 and 195, under the heading of "Diagnostic Criteria for Schizophrenia," a listing of most of the standard descriptions of psi faculties as, and I quote, "characteristic of psychotic symptoms". Some of the "symptoms" are given as thought broadcasting, superstitiousness, belief in clairvoyance, telepathy, the sixth sense, and claims the one can feel or others can feel one's feeling. Also alluded to are religious-type visions and ecstasies, and all of these "symptoms" are specifically characterized as, and I quote, "Odd beliefs or

magical thinking, influencing behavior and inconsistent with cultural norms."

More basically, these "characteristic psychotic symptoms" are generally grouped under the heading of "bizarre delusions," which are immediately defined in the Manual, as, and I quote: "involving a phenomenon that the person's culture would regard as totally implausible." In other words, one of the fundamental bases for determining whether something should be thought of as psychotic or not exists with whatever your culture would regard as totally implausible. Well, I am now old enough to have been born into that culture which generally believed and taught that men would never walk on the moon; and that the atom was so indestructible that its would never be split or its energy ever released. I distinctly remember, in my childhood, people speaking of such "bizarre delusions".

In fact, this criterion of "the person's culture would regard as totally implausible" appears to be almost the exclusive basis for such determinations. This was confirmed within our project at Stanford Research Institute where we possessed extensive capabilities, through the national interlibrary-loan system, of collecting all relevant research on the mind. At SRI we were naturally interested in discovering the clinical research data and documentation utilized to determine how and why psi faculties had been identified and confirmed as symptomatic of psychosis. We were especially ardent in this search, for we wanted to learn if we working on the project researching psi faculties were a group of psychotic lunatics. To be thorough and impartial, a good part of this documentation research was sub-contracted to Stanford University Medical Hospital, one of the most renowned in the United States.

The result of this gargantuan collection effort was this: what was called the "International Census of Waking Hallucinations in the Sane" was undertaken just before the turn of the century, and involved a survey of about 3,000 people in England and Europe who volunteered to admit that they had experienced some kind of hallucination.

It's important to note here that both the advance hypotheses and the preconceived model of this Census was that the phenomena were hallucinatory or illusory and were to be interpreted in no other way. The results of the Census, and its theoretical concepts, were ultimately made widely available in 1898 as Hallucinations and Illusions: a Study of the Fallacies of Perception, by Edmund Parish, published as a volume of the then enormously influential Contemporary Science Series.

Via this Census, it was determined that about less than 10% of the sane had temporarily experienced some kind of telepathic or clairvoyant hallucination, but in some categories of

experience the average was much higher. Immediately thereafter, several international congresses utilized this data, and it was generally determined that the less-than-10% figure indicated that the hallucinatory abnormalities were only very marginal and thus unimportant occurrences of the phenomena. But the direct result here should be pointed up. It's this: that psychical faculties were not judged to exist within 10% of the population, since the possibility of such faculties were excluded by the survey model; rather it was judged that only a minimal 10% of the sane population experienced hallucinations and illusions part of the time.

As the documentation research at SRI revealed, as of 1973 when our project there got underway, no other effort of any kind had been made to survey populations with regard to determining the frequency of occurrence among them of the allegedly psychotic, psychical, and parapsychological phenomena. Thus a 76-year epoch had passed, during which no scientific research had taken place either to substantiate or contradict the prevailing mainstream idea that psi faculties were hallucinatory and hence of psychotic etiology. It's also worth pointing up here that parapsychology never launched such a survey, when, in fact, doing so might have been expected of it.

In 1973, though, under the supervision of Dr. Andrew Greeley, larger national surveys were undertaken under the auspices of the University of Chicago's National Opinion Research Council (NORC). In 1985, the same research organization conducted larger additional surveys along these lines. As a result of the latter 1985 poll it was determined that 67% (two-thirds) of all adults reported having experienced various forms of ESP. These statistics were publicly presented in various forums and media outlets -- for example, in an article by Dr. Greeley entitled "Mysticism Goes Mainstream" published in American Health, Jan-Feb 1987. In 1989, the TV program 48 Hours, not content with the Chicago polls, conducted one of their own -- and turned up with similar and even more suggestive high statistics.

Thus, to let my sardonic nature vent itself here, if the American Psychiatric Association's Manual is to be utilized, then two-thirds of the American population polled by the Chicago University are psychotic.

Now, to return to the mainframe of this talk, public perception of psi faculties and general scientific perception of it are two different things. As I mentioned earlier, the justification in modern science for the rejection of psi was that no biological or organic basis for it existed -- and until such a basis was discovered, psi phenomena were to be considered products of psychological disorientation, the judgmental basis of which has now been established by what I've just reviewed briefly. It's worth pointing up here that the absence of a

biological basis for ESP, and so forth, accounts for why the phenomena were considered to be only psychological. Hence the term Parapsychology -- instead of Parabiology.

Public perception of the existence of psi has little meaning here, and in fact does not inspire scientific mind-sets to alter either their venues or purviews. For example, it is now 1994, and the leading science magazine Nature (published in England) follows its established editorial policy of automatically rejecting any scientific papers submitted to it involving parapsychology or psi phenomena. Scientific American follows suit, as do most other leading scientific periodicals and publications. In a descriptive sense, then, psi and parapsychology are still "ghettoized", in the sense that they represent what the American Psychiatric Association's Manual says they do: "bizarre delusions involving phenomenon that the person's culture would regard as totally implausible."

The continued rejection in these terms, or any other, is exceedingly strange -- because, as I'll review shortly, biological and organic bases for at least several extraordinary human perceptual faculties commonly classified as psi in parapsychology have been discovered. And they have been discovered not by parapsychologists, but by neurophysiologists, neuroanatomists, and biopsychologists -- and which fields are perfectly respected in science in general.

But this leads now, as it did during the 1970s, to which or what culture is being utilized to determine misfitting implausibilities. If, as I will show, and others have already announced, a bio-organic-causative basis does exist for some psi faculties, then the rejection of psi has having no biological basis is to be voided or considered fulfilled and resolved. Some aspects of the discovered biological basis for certain psi faculties have existed since the 1950s -- certainly enough time for the implications to have penetrated even remote corners of mainstream science and media coverage. Yet this positive shift has not taken place in mainstream culture, which apparently remains recalcitrant toward the discoveries regarding the biological basis for psi.

Now, "mainstream culture" is not really a culture of and in itself; rather it consists of a number of sub-cultures that integrate enough so as to have attitudes and expectations in common. How many of you here know what a Chinese Box is? You open one box only to find another one inside of it. When you open the second one you find a third, and so forth.

Here is one of these endless boxes. If bio-bases for certain kinds of psi have been discovered, and if science in general still rejects the fulfillment of its own demands in this regard, then something else is crucially involved. A clue to

what this "something else" might be was somewhat inadvertently revealed behind the scenes of general public information, and more or less kept concealed by lack of media-outlet interest.

During 1986 and 1987, a formal debate between skeptics and parapsychologists was arranged, principally under the auspices of Drs. K. Ramakrishna Rao and John Palmer on the parapsychologists side, and Dr. James E. Alcock, a psychologist and leading scientific skeptic, on the skeptics' side. The way the debate was formulated was that the parapsychologists would gather a series of papers from a number of parapsychologists showing statistical evidence that psi existed.

In reply, the skeptics were to present a series of papers commenting on what they thought to be inadequacies of the parapsychologists, and showing that psi didn't exist. All the papers were to be studied by a third independent refereeing group which was to compare the two bodies of papers and see what could be made of them. The whole of these papers, along with the commentaries of the third refereeing group, were then published in toto by the Cambridge University Press in Behavioral and Brain Sciences (1987), pages 539-643.

The parapsychologists' papers were grouped under the general heading of "The anomaly called psi: Recent Research and Criticism." In the "Abstract" heading the parapsychologists' papers, they offered that statistical evidence for psi was good but not enormous, but that the statistical samples were consistent enough to lead them to feel that "a wide range of research seems to converge on the idea that, because ESP 'information' seems to behave like a weak signal that has to compete for the [converging] information-processing resources of the organism ... that such a meaningful convergence of results suggests that psi phenomena may represent a unitary, coherent process whose nature and compatibility with current physical theory have yet to be determined."

More simply put, this means that psi information signals exist within the noisy information-processing resources of our organisms, and that a reduction of the noise may permit psi signals to emerge more clearly.

Now, in response to the parapsychologists' papers, presented under the heading of "The anomaly called psi: Recent Research and Criticism," the skeptics produced their answering collection of papers under their heading of "Parapsychology: Science of the anomalous or the search for the soul?" In part, the "Abstract" of the skeptics' papers reads: "Although many parapsychological research projects have been carried out under what have been described as well-controlled conditions, this does not by itself make a science, for unless and until it can be demonstrated that paranormal phenomena really exist, there is no subject matter

around which a science can develop. .. It is argued in this paper that parapsychological inquiry reflects the attempt to establish the reality of a nonmaterial aspect of human existence, rather than a search for explanations for anomalous phenomena."

Now, I've worked full-time for twenty years in the stygian gloom of parapsychology laboratories of all kinds. I've many criticisms to voice as many parapsychologists have found out, mostly because I can't keep my big mouth diplomatic enough. But I've no criticism here concerning their alleged search for the soul -- simply because I can't remember hearing any of them use the term, and I've very seldom seen it in print in the entirety of psychical research and parapsychological archives.

As a matter of real fact, the accumulating of statistics under well-controlled conditions is the accepted mainstay of the scientific process. It's the accumulating of the statistics that gives evidence of something existing; and in all other fields except parapsychology, even statistics of lesser minute degree are automatically taken as evidential.

Although I probably don't need to point out to your excellent qualities of understanding, both the parapsychologists and skeptics brought the best they could offer in the way of eminence and expertise to this debate. The parapsychologists presented statistical evidence: in response the skeptics presented a philosophical (or anti-religious) argument about the soul. It was clear to the independent third-party referees that the two groups weren't talking about the same or even similar substantive matters, and most of them said as much in print -- after which the documents published in the esteemed Behavioral and Brain Sciences issue were ignored by media outlets and thus settled into the cultural oblivion where implausibilities are urged to go and not bother anyone.

In fact, two years before they organized their curious debate, if both the parapsychologists and the scientific skeptics had read a book that came out in 1984, and was easily available via the esteemed publisher, Simon & Schuster, both sides might have approached the facts and issues involved quite differently.

Now, having dragged you all through the foregoing, let me begin to readjust our cultural perceptual understandings, psychic or otherwise, about this. This won't take long, and is actually quite simple.

The term parapsychology came into use because no biological bases for psi had been discovered, or were even thought to be plausible by the socially prevailing opinions of the hard sciences. Hence, psi faculties were ascribed to the soft sciences as being exclusively mental or psychological in origin. But the socially prevailing opinions of the soft sciences didn't

feel that the faculties were plausible, either, and thus held them to be hallucinatory. So the term Para (meaning "beyond") Psychology was adopted -- this a term never accepted as meaning anything real in the soft sciences, but accepted as the illusory matter of scientifically rejectable parapsychology only.

Upon these grounds, then, or the lack of them, both the hard and soft sciences of the modern epoch rejected as culturally implausible the real existence psi faculties in the bio-psychological human -- sometimes expressed as the mind-body human. Thus, we are dealing here with the two mind-sets of the hard and soft sciences, both of which rejected psi faculties as being, in their terms, culturally implausible. I'll be brave enough to point out that the socio-cultural rejection of psi by both socio-mind-sets rests on an almost total vacuum of direct research by their own functionaries and representatives -- albeit with one novel and even shocking exception, to which I'll allude at the end of this talk.

The new and now on-going discoveries of biological bases for many psi faculties is now best expressed, perhaps, as Parabiology, meaning beyond former standard concepts of biology. Or, perhaps, the term Parapsychobiology -- which, if translated into Russian, would become the term translated back into English as bio-communications or bio-information transfer.

The new discoveries regarding the biological bases for psi faculties so far roughly fall into five categories. I have to get a little technical here, but I'll simplify just ahead. These five general categories are: minute chemical receptors and sensors; minute chemico-electro receptors and sensors; neural-network exchanges of information in the bio-internal body substrates; bio-electromagnetic information receptors and sensors; and bio-information transfer networks at the atomic, molecular, and neurological levels.

If these new terms are somewhat confusing, well don't worry too much. What they amount to is the confirmed scientific understanding, in the novel hard sciences, that the human bio-body possesses at least seventeen delicate systems of receptors and sensors that can integrate with our normal five, and, given a chance, can also integrate with our mental cognitive powers -- to result in, yes, what have otherwise generally been called "psi faculties."

Now, to give you here some broad idea of the cutting-edge sciences involved, I'll quickly read through a list of 21 of them:

- Electro-chemical physiology
- Neurobiology
- Neurophysiology
- Neuropsychology

Bio-radiation studies
 Hormone and Hormonal transmission research
 Chemical signal research
 Bio-electric research
 Brainwave research
 Bio-sensitivity research
 Bio-electric information transfer research
 Sensory coding research
 Bio-magnetic navigation research
 Bio-electronic systems research
 Bio-electric field detection research
 Electrophysiological studies
 Pheromone and pheromone transfer research
 Multi-stability in perception research
 Subliminal perception research
 Neuro-magnetic response research
 Bio-infrared and bio-ultraviolet perception research

At this point, I could adumbrate upon many more than just a thousand scientific papers about these discoveries published in science literature, even in the esteemed leading science periodicals such as Nature and Scientific American. The authors of these papers, though, never use the terms psi, psychic, or parapsychology, since the mere introduction of them would cause their papers to be rejected. The editors and peer-review systems of such publications apparently don't realize, for example, that "bioinformation transfer over distance" means about the same thing as "telepathy" and/or "clairvoyance". But this is merely part of the Particular Situation I referred to at the beginning of this talk.

I now hold up in my hand before you, so that you can see that it really exists, a book published in 1984 via Simon & Schuster, by Robert Rivlin and Karen Gravelle. This book is complete with bibliography of scientific sources, but is easy to read. It's entitled, somewhat misleadingly, as Deciphering the Senses: The Expanding World of Human Perception. It should have been called something like The Discovery of the Biological Basis for Psi and Other Anomalous Perceptions. The discoveries brought together and presented in this book, although in popular style, are based on hard scientific discoveries that have been achieved in other disciplines outside and independent of parapsychology; and as such the sum of them clearly fulfills -- at least regarding certain forms of psi -- the earlier scientific demand that a bio-organic basis for psi be discovered.

I quote from the fly-leaf: "For centuries we have used an oversimplified and inaccurate model to explain the human senses. Even now, high school biology classes still teach the 'five senses'. But recent scientific research has discovered that there are many more than five senses, and these discoveries have radically changed our understanding of what the senses are and

how they work. Rivlin and Gravelle redefine for the general reader the spectrum of human perceptions from the normal to the newly discovered to the extra-sensory." As chapter one indicates, the book discusses "The seventeen senses" additional to our usual five ones, and then goes on to place the newly-discovered senses in context with the usual five. They consume eight chapters to do so before they reach chapter 9, entitled "Extra-sensory perception."

Now, lest there be some misunderstanding here, the two authors are not describing psychological or mental functions. They are discussing the actual existence within our biology of minute physical-chemical-bioelectromagnetic "receptors" and "sensors" that interact within networks of the "information-processing resources of the organism" as hypothesized by the parapsychologists in their debate with the skeptics. As is pointed out, five senses are obviously not enough to account for the huge range of sensory possibilities of which the human species is capable, while seventeen senses is probably a more accurate count, with more probably yet to be discovered.

As to what these seventeen new senses are. The seventeen new senses interact with each other to provide a rather extensive list, many of which have in the past been referred to as "psychic." Since we don't have time here to go through them, I've brought a few copies of that list to hand out, along with some copies of this lecture, and all of which you are free to duplicate.

But, for example, the bio-body is now known to have a functioning vomeronasal system containing receptors enabling, at the bio-subliminal level, the detection of minute amounts of chemical signals that tell us about another's sexual receptivity, fear, anger, and other emotions -- an aptitude more commonly referred to as "psychic vibe-sensing."

In another category, through the use of a newly invented device called the SQUID, scientists can and have measured and begun to classify the brain's electrical activity outside of the scalp, which in turn has led to discoveries that bio-electric activity extends to some distance beyond the skin, which in turn has led to the discovery of bio-electric sensors not only in the skin, but in the neuropeptide activity that transmits all kinds of subtle senses information through the immune system and into the brain and back again into the body's extremities and all its internal organs, including into its surrounding bio-electromagnetic field.

Now, discovered bio-electromagnetic fields extending outside the scalp and outside of the skin clearly equate to the "auras" that many clairvoyants have specialized in "seeing." Rivlin and Gravelle even hypothesize that, and I quote, "thoughts may,

indeed, have wings, and some of us may have the ability to sense what others are thinking," via these newly discovered bio-electromagnetic receptor-sensing networks. The two authors wonder: "Do some psychics and mystics have this ability, vastly magnified, so they can sense the electricity from considerable distances?"

Actually, this is the same question that those researching electromagnetism and bio-electromagnetism have been wondering about for over fifty years. So it's worth pointing up here that the existence of extensive bio-electromagnetism essentially was demonstrated late during the last century, but its existence has not figured very much either into scientific psychology or in scientific parapsychology -- both of these two fields having managed mutually to ignore it altogether.

Did you know that in addition to yourself being a meat body with eyes, livers, hearts, and appetites of various kinds, you are also a bio-electronic one? Have you ever thought of yourself as such? If you begin to, well, some interesting might begin to happen. [REDACTED] is one of the leading researchers in the United States regarding electromagnetism and bio-electromagnetism. With [REDACTED], he published, in 1985, a book entitled The Body Electric: Electromagnetism and the Foundation of Life (William Morrow, New York), and which "tells the story of our bioelectric selves." A companion book is Harold Saxton Burr's Blueprint for Immortality: The Electric Patterns of Life (Neville Spearman, London, 1973-1988).

Bob Becker has made many unequivocal statements regarding the psi implications of bio-electromagnetism. For example, as published in Psychoenergetic Systems, 1977, Vol. 2, pp. 189-196, in an article entitled "An Application of Direct Current Neural Systems to Psychic Phenomena," he states that "The concept of a primitive electronic communication system in all living things can be a useful tool in understanding both 'normal' and 'paranormal' phenomena that have lacked a rational biological explanation. Indeed, it appears that human beings are tied to the universe in a web of electromagnetic energy."

At this point, I believe I've now presented for your consideration the rudiments of the Particular Situation I referred to here at the beginning. The Particular Situation consists of three facts: (1) science demanded that a bio-organic explanation for psi faculties be found before it could accept them as real; (2) bio-organic explanations have been found for many kinds of psi faculties; and (3) everyone seems to be ignoring both the facts and the implication of (2) as just stated. Well, not exactly "everyone." But certainly academia and media, even the venerable Time magazine, so far -- which, I predict, will one day ahead publish a cover-story advising: "Surprise! Everything you thought you knew about psi is wrong."

As to more of what the seventeen senses are. Did you know that the soles of your feet and the palms of your hands contain minute magnetic receptors and sensors that "recognize" minute and gross changes in local magnetism? Here are the rudiments of dowsing, healing, and various rough forms of psychometry which means psyching-out what something is by merely holding it. Alas, though. If you haven't built neural pathways to your cognitive faculties, you probably won't be able to sense what the soles of your feet are. Of course, I could extend this lecture indefinitely and outline why most of us haven't built such neural-cognitive pathways -- largely because we don't want to seem implausible in our culture, do we?

In coming to the end of my allotted time here, I'd like to report to you an astonishing event that's about to occur. I've referred several times to the venerable third revision of the Diagnostic and Statistical Manual of Mental Disorders published by the American Psychiatric Association, and which refers on page 195 to "superstitions" as being at least schizophrenic in origin, and which goes on to include, and I quote: "unusual perceptual experiences such as sensing the presence of a force or person not actually present." Among other things this is taken to refer to are religious and spiritual problems that entail experiencing clinical or near-death episodes in which disembodied or non-physical forces and persons are encountered, and then coming back from them to life.

In its February 10, 1994 pages, the venerable New York Times reports that the APA is about to publish a fourth revised edition of its Manual. In it will be a new category of interest, which is, and I quote, "In a change intended to encourage mental health professionals to view a patient's religious experience more seriously, the next edition of the standard guide for psychiatric diagnoses includes a new entry entitled 'Religious or Spiritual Problem'." The Time's article goes on to state that in the past "such problems were being misdiagnosed and mistreated" and then goes on to state that "researchers now recognize that near-death or after-death experiences are not uncommon, but that in the past psychiatrists might have mistakenly diagnosed them as a symptom of a mental disorder." An article published in The American Journal of Psychiatry is quoted as saying, and I quote it here, too, "that the stereotyping of religion as clinically harmful ... contradicted the extensive findings that religion and/or spiritual experiencing was positively associated with mental health."

Well! It appears that the skeptics of parapsychology have a new forum in which to discuss the soul, don't they, and this forum doesn't any longer consist of timid parapsychologists, but which is no less than the esteemed American Psychiatric Association.

Upon this sardonicism, I now rest my case before you, or at least rest whatever you've been able to make of it. I hope I haven't bored you to death, and I thank you for your attention.

A PARTIAL LISTING OF
NEWLY-DISCOVERED HUMAN "PSI" SENSES
NOW KNOWN TO HAVE BIO-ORGANIC BASES

The normal 5-senses are: seeing, hearing, feeling, tasting, smelling. Note that the extent of these differ from individual to individual, and because of cultural-environmental reasons.

Bio-organic bases have been discovered for the following additional senses:

- Receptors in the nose that "smell" emotions, and that can identify motives, sexual receptivity, antagonism, benevolence, etc. (Psi vibe-sensing).
- Receptors in the ear that "hear" differences in pressure and electromagnetic frequencies. (ESP-like).
- Skin receptors that detect balance and imbalance regarding what is exterior to the bio-body. (Remote-sensing, a form of ESP).
- Skin receptors that detect motion outside of the body, even when the body is asleep. (A form of ESP).
- Directional-finding and locating receptors in the endocrine and neuropeptide systems. (Directional dowsing, a form of ESP or intuition).
- Receptors that identify "fluid" motions of horizontal, vertical, diagonal, even if not visually perceived. (As in the psi portion of the martial art Akido.)
- } Subliminal sense of pitch of sound: }
- } Subliminal sense of heat: }
- } Subliminal sense of frequencies }
- and waves, either mechanical }
- or energy: }
- Sense of non-visual wave motion. (A form of ESP).
- Sense of non-visual oscillating patterns. (A form of ESP).
- Receptors that identify positively- or negatively-charged particles at the atomic level. (Psi equivalent is "micro-psi", rare or otherwise unidentified, except in the case of [REDACTED] (the famous Theosophist and clairvoyant) who published Occult Chemistry (1908, 1918), who correctly described sub-atomic particles, many undiscovered at the time, but discovered since. Micro-psi faculties are mentioned as one of the "sidhis" in the Yoga Sutras of Patanjali.)
- Remote-sensing systems. (Forms of clairvoyance.)
- Skin receptors that "recognize" the temperament of other biological organisms. (A form of psi "reading").
- Eidetic senses -- seeing images after their source has been removed from the visual field. (A type of clairvoyance.)
- Sense of magnetic fields }
- Sense of infrared radiation} (All commonly thought of as forms
- Sense of electrical energy } of ESP).

Sense-transducing of various forms of mechanical, chemical, and electromagnetic energy into meaningful nerve impulses. (All commonly thought of as forms of ESP).

Whole-body receptors (hundreds of different kinds of them) to detect pheromones, sexual receptivity, fear, love, admiration, danger, pain in others. (All forms of ESP).

Receptors that sense gravitational changes. (Dowsing.)

Neurological sense of appropriate and inappropriate food substances. (A substrate for psi healing.)

Neurological senses for interpreting modulated electronic information by converting them into analog signals for mental storage, interpretation, and cognition. (One of the bio-bases for telepathy.)

Bio-electronic receptors for sensing radiation, including X-rays, cosmic rays, infrared radiation, and ultraviolet light, all of these receptors being found in the retina of the eye. (Psi equivalents not known, and only have been researched in the bio-communications model of the former Soviet Union and the People's Republic of China.)

Sense-receptors for local and distant sources of heat. (An unnamed psi aptitude, but one familiar to Amerindians.)

Receptors that respond to exterior electrical fields and systems. (The basis for "aura" perceiving.)

The following are generally referred to in the new sciences as "Human Semaphore Capacities":

Skin receptors for sensing perceptions of bonding or antagonism. (Thought of as forms of intuition.)

Senses for "languages" of nonverbal sensory communicating. (Usually grouped with "vibe-sensing.")

Receptors that trigger alarm and apprehension before their sources are directly perceived. (A form of psi foresight or foreseeing or intuition.)

Combined senses (neural networks) for making meaning out of 130 identified nonverbal physical gestures and 20 basic kinds of nonverbal messages. (Sometimes thought of as intuitional character assessment, and as a psi faculty.)

Senses for registering nonverbal emotional waves. (Another form of intuition.)

It is now known, for sure, that the PINEAL gland, if in good working order, responds to:

Cycles of light and darkness, anticipating them with accuracy as the daily motion of the sun and moon change. (A kind of foreseeing.)

Solar and lunar rhythms, solar disruptions (flares) and moon-caused tidal changes (water or geophysical ones), and can sense "coming" earthquakes and storms. (A form of predictive ESP, especially noted in farmers, but also in cows, dogs, cats, and snakes.)

Geo-electromagnetic "pulses". (Psi equivalent unidentified).
Magnetic fields, especially biological ones. (Aura-sensing).
If the pineal is fully functional, it acts as a nonvisual photo-
receptor. (Psi equivalent is sometimes thought of as "X-ray
vision").

Please note that this list is not complete.

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